PREPARED BY ADIL ALVI

UNSOLVED FINAL TERM PAST PAPER OF MTH101

Aug 28, 2019

One question is sigma notation
One is to find the arc length
One question is definite integral and integration by substitution method
One is Taylor series
Jo last five chapters they us mein se zyada mcqs aye thay so unhon ne question type e mcqs diye thay

Feb 22, 2018

Last 6 lecture achy sy parh lyna... 5 marks k questions a rhy ha
Total question are 52
Convergence and divergence test, (imp) *
Series and sequence, (imp) *
Limit,
Newton's method,
integration (imp) *
Derivatives (imp) *
40mcqs,

Aug 20, 2016

Q1=3+ + +…..+
Q2=f (t) =2t-1 find out the increasing and decreasing point
Q3= evaluate the sum
Q4= evaluate it
Q5=series 1+4+9+…….100 written this series in sigma notation
Q6= evaluate the limit.
Q7= evaluate the improper integral.
Q8= evaluate improper integral
Q9= make the 4 subinterval of this and find x*k which is as left end point (is Ki statement sahe sy yad ni ha tha aysay he kuch)
Q10= evaluate the integral

For any query feel free to ask adilalvi21@gmail.com

PREPARED BY ADIL ALVI

2015

Q#1. Find an equation of the line y-intercept -7 and slope -5.

Q#2.Find the definate integral indicating the arc length of the curve y=2x from x=0 to x=1, But don't evaluate it further.

Q#3.Evaluate by using the subsitution methode.

$$\int\left(1-\tfrac{1}{t}\right)\cos(1-\ln t)\,dt$$

Q#4.Use the Subsitution methode to express the definate integral in term of the variable 'U' but do not evaluate.

$$\int_0^{\pi} e^{\cos t}\left(\sin t\right)dt$$

Q#5.Evaluate using first fundamental theorem of calculus.

$$\int_0^1\left(x^5-x^3+2x\right)dx$$

Q#6. If F is a smooth function on [1,3], then find the definate integral indicating the arc length of the curve $x = y^{\frac{2}{3}}$ **from y=1 to y=3.**

Q#7. Evaluate:

$$\lim_{y\to\infty}\frac{y^3+2y^2-5y+3}{y^2-3y+1}$$

Q#8. write this in Sigma Notation.

$$3\times1+3\times2+3\times3+.....+3\times20$$

Q#9. If $F(x)=\int_0^x(\sin t)\,dt$ **then find F(x) by 2nd fundamental of calculus.**

Q#10. Find the vertical asympototes

$$f(x)=\frac{x+5}{x^2-9}.$$

Q#11. Use L' Hopital Rule

$$\lim_{x\to\pi}\frac{\sin^2 x}{\cos 3x+1}$$

For any query feel free to ask adilalvi21@gmail.com

PREPARED BY ADIL ALVI

2014

Evaluated limit 5marks

$$\lim_{x\to\infty}(\sqrt{x^2+5x}-x$$

Find integral using first theorem of calculus 5marks

$$\int_0^1 (x^5-x^3+2x)dx$$

An convergence or divergence find 3marks

$$a_n=\frac{\sqrt{n+1}}{n}$$

Find rate of change at an arbitrary point of f 5marks

$$f(x)=\sqrt{x+3}$$

Find limit 2marks

$$\lim_{x\to 5} h(x)$$

$$h(x)=\sqrt{2x-1}$$

Find the spring constant 'k'; if a force of 10N is required to stretch a spring from its natural length of 4.8m to a length of 6.8m? 5marks

aik example ay thi lecture no.38 page 123 W=FD 5marks

For any query feel free to ask adilalvi21@gmail.com

PREPARED BY ADIL ALVI

2013

Evaluate the improper integral $\int_3^{+\infty} \frac{dx}{2x^2}$ 2

Explaine alternating series test..... 2

find $\frac{dy}{dx}$ at (0,1) if $xy^3 = 54$.....2

divide the interval [2,6] into (4) subinterval to find $f(x_k^*)$ for the line $y = 5x + 1$ when x_k^* is right end point o each subinterval 2

use the subsitution method to expresse the definite integral in terms of the variable "u"

but do not evaluate $\int_0^1 t\sqrt{1-t^2}\,dt$.....3

show that series $\sum_{k=2}^{\infty} (-1)^{k-1} \frac{2^{k-1}}{(k-1)!}$.....3

find the spring constant "K", if a force of 10N is reqiured to stretched ∞ spring from its natural length of 4.8 m to a length of 6.8 m 3

if the point (-1,2) lies on the circle with center (2, -3),
the find the radian of the circle.....3

evaluate the integral : $\int_{\pi}^{x} \sin 2t\,dt$ 5

the regin bounded by x- axis, the graph of the equation $y= x^2$
and the line x= 2 is revolved a about x- axis.
find the volume of the resoluting solid 5

find the radius of convergance for the following power series

$\sum^{\infty} x^{2x}$ 5

Evaluate $\text{Lim}_{\theta \to 1} \frac{\theta^4 - 1}{\theta^3 - 1}$.....5

For any query feel free to ask adilalvi21@gmail.com

PREPARED BY ADIL ALVI

Feb 24, 2013

Mera papr ziada 40 to 43 lec se mcqs thay.
1 question slope ka tha (5)
1 question volume ka tha (5)
1 was from integration (5)
1 question from chan rule (2) marks

May 20, 2012

Q1) Find the point of discontinuing of the given function: f (x) = 3x + 1 X2-7x+12? 2 Marks
Q2) Find dx at (-5, 6) if x2 + y2 =0 2 Marks
Q3) Determine whether lim x−>0 1 – cos |x| exists or not? 3 Marks
Q4) if f '(t) = 2t then find the intervals on which the function f (t) increasing and decreasing? 3 M
Q5) Use an epsilon-delta argument to prove that lim x−>0 (3x + 5) = 35? 5 Marks
Q6) if f(x) = x -5x+6 then find the intervals where the function is concave up and concave down? 5 Marks

Feb 4, 2012

41. Find a definite integral indicating the area bounded by the curves y = x – 7 and 2 y = x between x = 0 and x =1. But do not evaluate the integral. 2 marks
42. Evaluate the improper integral 2 marks
43. Only write down the McLaurin series for Cos(x) 2 marks
44. Evaluate 2 marks
45. Show that lim 0 k for the following alternating series
46. Evaluate the integral: 3 marks
47. Find the work done by the force 400x if an object moves in the positive direction over the interval [0.15, 0.18]? 3 marks
48. Find equation of the line having slope 3 and passing through the origin. 3 marks
49. Evaluate the integral:
by using a proper substitution.5 marks
50. Evaluate the improper integral5 marks
51. Show that Is divergent for the following alternating series.5 marks
52. Find the instantaneous rate of change of f (x) = x +1 at x 5 marks

Jul 21, 2011

40 marks objective
4 q of mark 5
4 of 3 mark
4 of 2 mark
slop find 5 marks
n!/1o^n tell the converge of diverge 5 mark
find the area 5 mark
find surface 3 mark
evaluate integral 3 marks
find the region 5 marks

For any query feel free to ask adilalvi21@gmail.com

find the improper integral 3 marks
differentiate the equation 3 marks

Feb 18, 2011

40 mcqs
12 long and short questions
2 questions were from simple derivation
3 questions were for finding volume
1 question were for l hospitals rule.
And remaining all questions were from integration

Feb 23, 2011

* Find integration by substitution: integration (1+square root(x)) 3/under root (x) (5)
* find d2y/dx2 of xy=x+y (5)
* one question is sigma notation (5)
* one is about finding volume around y-axis (5)
* solve inequality (2)
* integration 0 to 4 of 2x (2)
* integration 0 to 2pi of sin x (3)